

# Chapter5: DAC and ADC Part-II

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# Objective

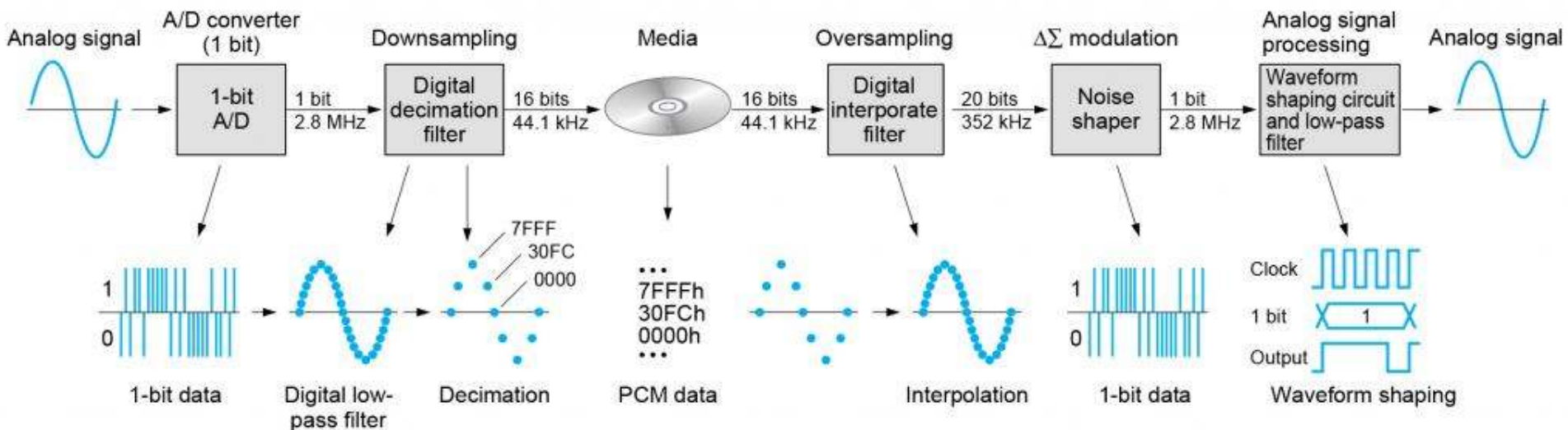
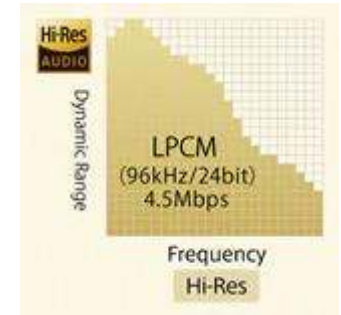
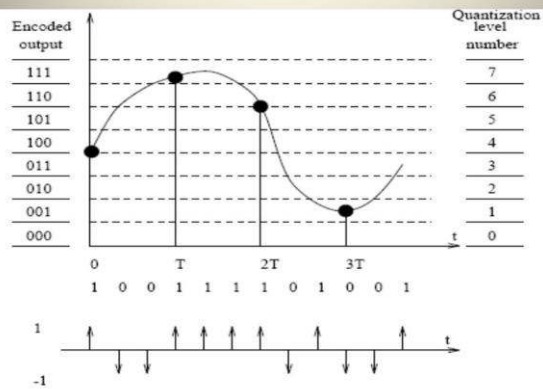
- To recognize mechanism of conversion from analog signal to digital signal, with digital circuit.
- To explain a R2R circuit to convert digital data to analog signal.
- To understand concept of convert analog signal to digital signal and digital to analog signal.

# Topic

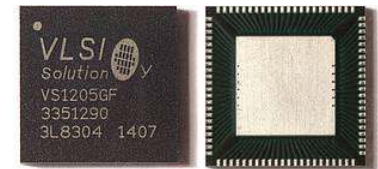
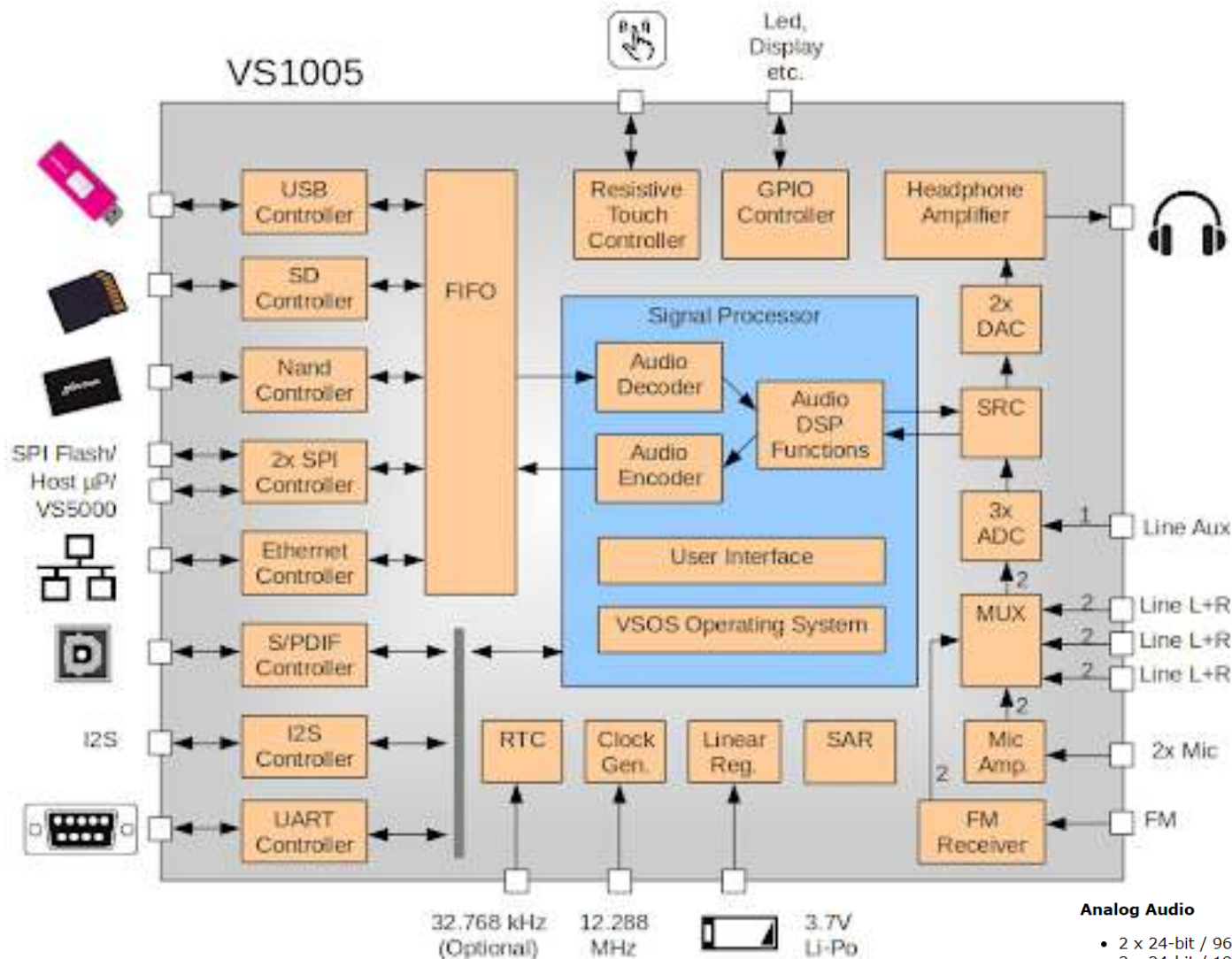
- Digital to Analog Conversion
  - R2R ladder circuit
- Analog to Digital Conversion
  - Flash ADC
  - Successive-approximations register (SAR)

# Overview

## Pulse Code Modulation (PCM)



# MP3 Signal Processor

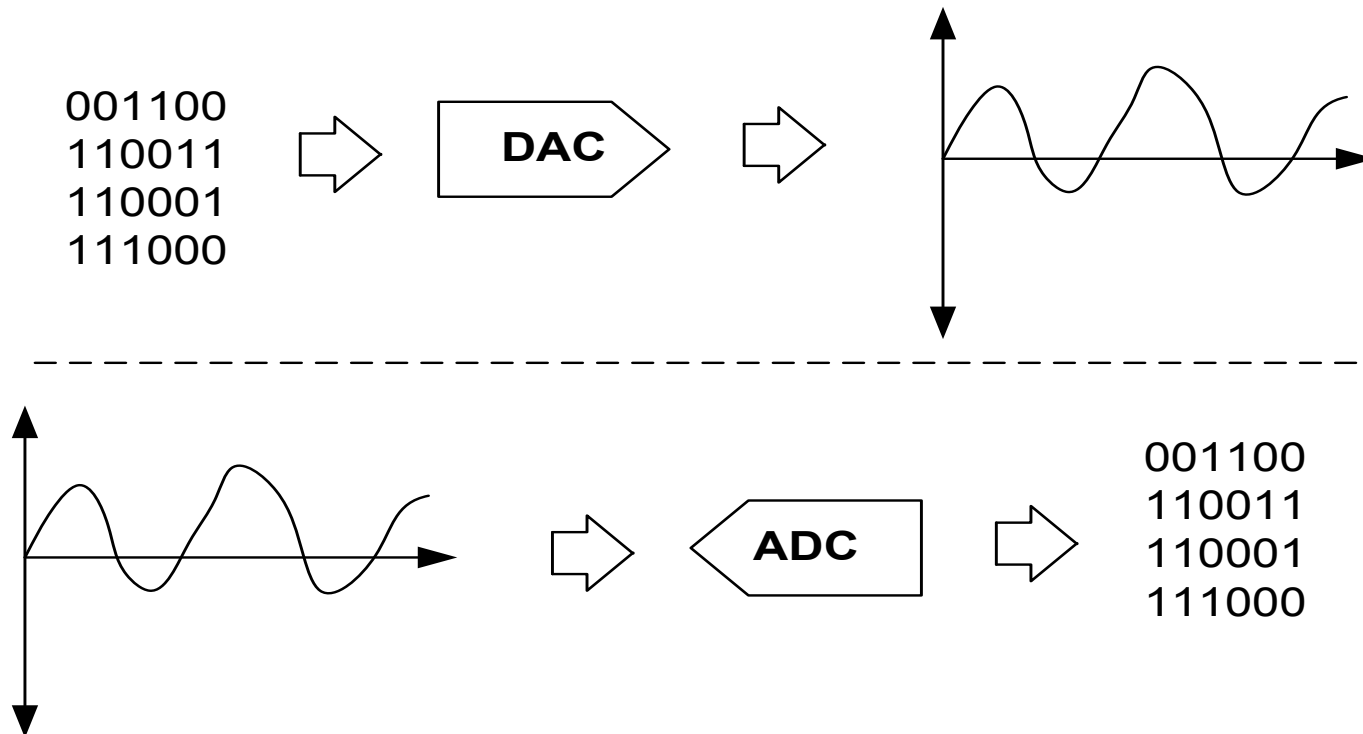


## Analog Audio

- 2 x 24-bit / 96kHz DAC
- 3 x 24-bit / 192kHz ADC
- Integrated FM tuner with RDS, Japanese band included (76-108 MHz)
- Stereo headphone output
- Stereo microphone amplifier

# Signal conversion concept

- DAC (Digital to Analog Conversion)
- ADC (Analog to Digital Conversion)





# **DIGITAL TO ANALOG CONVERSION**



# Example DAC design

Supposed

- 4-bits system
- Maximum output voltage at 12v



# Example DAC design

$2^4 = 16$  Levels (0 to 15)

$D_0 D_1 D_2 D_3$

Supposed

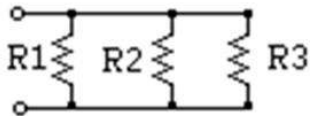
- 4-bits system
- Maximum output voltage at 12v

| D3 | D2 | D1 | D0 | bitvalue | Voltage |
|----|----|----|----|----------|---------|
| 0  | 0  | 0  | 0  | 0        | 0       |
| 0  | 0  | 0  | 1  | 1        | 0.8     |
| 0  | 0  | 1  | 0  | 2        | 1.6     |
| 0  | 0  | 1  | 1  | 3        | 2.4     |
| 0  | 1  | 0  | 0  | 4        | 3.2     |
| 0  | 1  | 0  | 1  | 5        | 4       |
| 0  | 1  | 1  | 0  | 6        | 4.8     |
| 0  | 1  | 1  | 1  | 7        | 5.6     |
| 1  | 0  | 0  | 0  | 8        | 6.4     |
| 1  | 0  | 0  | 1  | 9        | 7.2     |
| 1  | 0  | 1  | 0  | 10       | 8       |
| 1  | 0  | 1  | 1  | 11       | 8.8     |
| 1  | 1  | 0  | 0  | 12       | 9.6     |
| 1  | 1  | 0  | 1  | 13       | 10.4    |
| 1  | 1  | 1  | 0  | 14       | 11.2    |
| 1  | 1  | 1  | 1  | 15       | 12      |

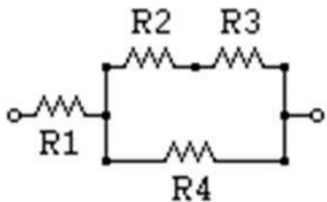
# DAC: Resistance and voltage divider



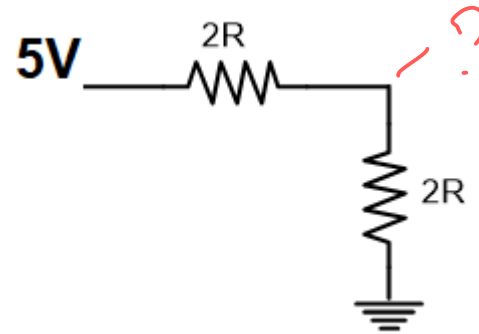
$$R_S = R1 + R2 + R3$$



$$R_P = \frac{1}{\frac{1}{R1} + \frac{1}{R2} + \frac{1}{R3}}$$



$$R = R1 + \frac{1}{\frac{1}{R2 + R3} + \frac{1}{R4}}$$

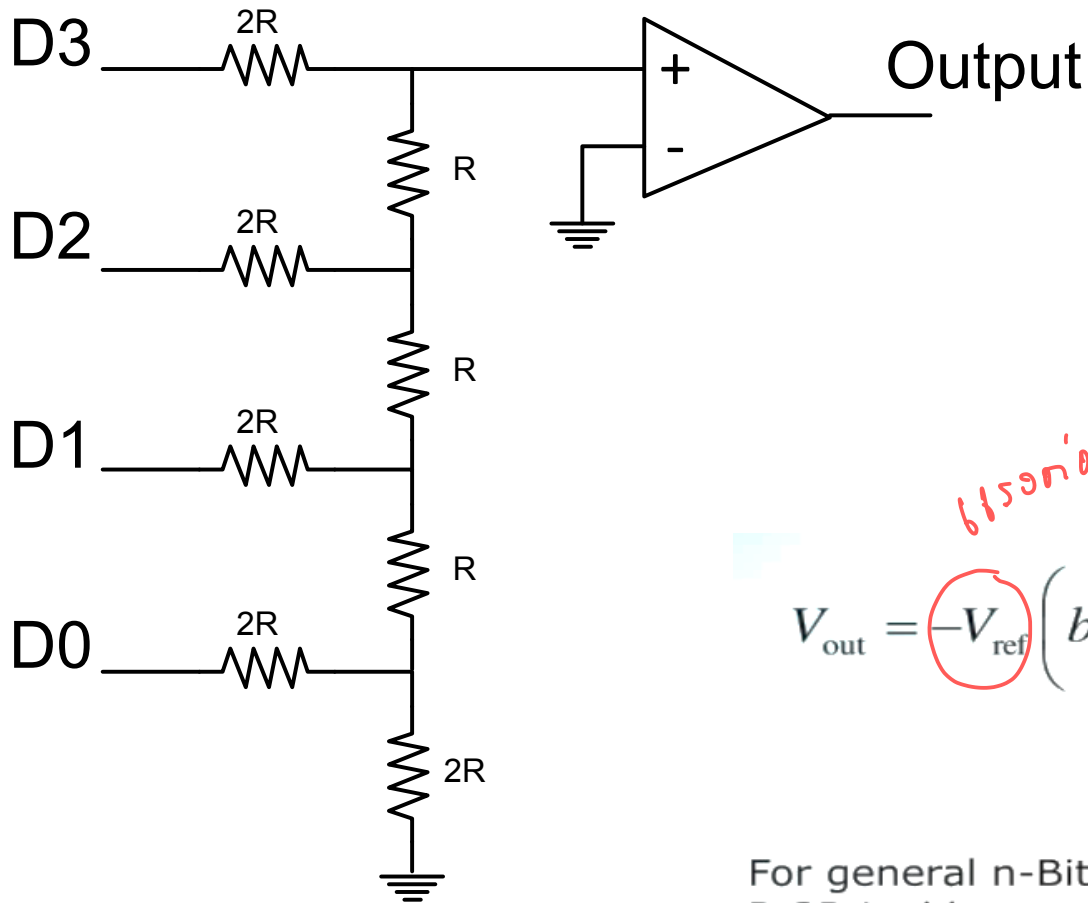


$$R = 100 \Omega$$

$$I = \frac{E}{R} = \frac{5}{200} = 0.025 A$$

$$V_x = I \times R = 0.025 \times 100 = 2.5 V$$

# DAC circuit (R2R)



*biweighting*

$$V_{\text{out}} = -V_{\text{ref}} \left( b_3 \frac{1}{2} + b_2 \frac{1}{4} + b_1 \frac{1}{8} + b_0 \frac{1}{16} \right)$$

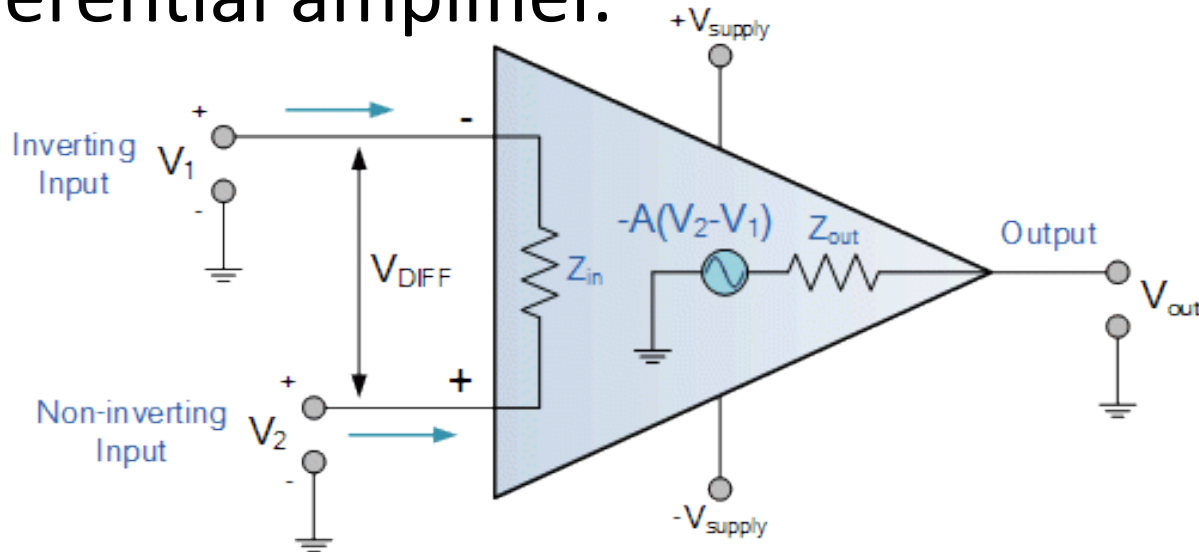
For a 4-Bit R-2R Ladder

For general n-Bit  
R-2R Ladder  
Binary Weighted  
Resistor DAC

$$V_{\text{out}} = -V_{\text{ref}} \sum_{i=1}^n b_{n-i} \frac{1}{2^i}$$

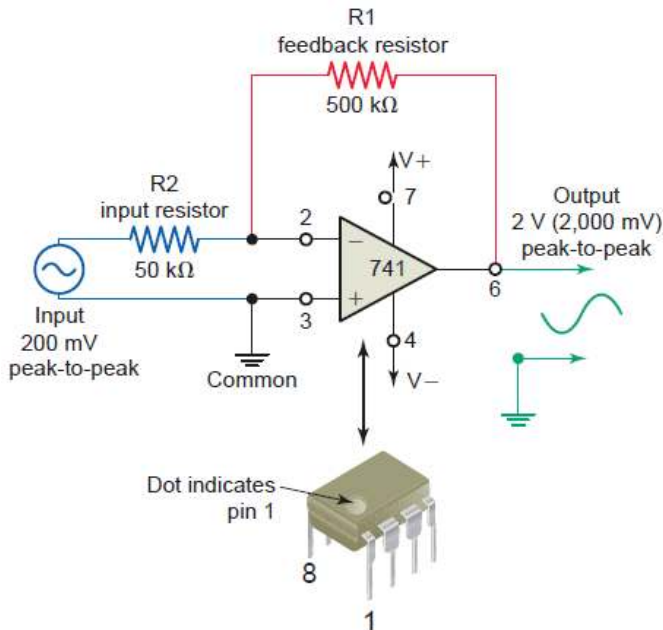
# Opamp

- Operational amplifiers is an IC amplifying a small signal.
- Opamp has a very-high input impedance and having two input polarities (+) and (-).
- With input characteristic, Opamp uses for differential amplifier.



# Opamp in the market

- Famous IC code
  - LM324
  - LM741
  - AD822



## Single-Supply, Rail-to-Rail Low Power FET-Input Op Amp AD822-EP

### FEATURES

- True single-supply operation
  - Input voltage range extends below ground
  - Output swings rail-to-rail
  - Single-supply capability from 5 V to 30 V
  - Dual-supply capability from  $\pm 2.5$  V to  $\pm 15$  V
- High load drive
  - Capacitive load drive of 350 pF,  $G = +1$
  - Minimum output current of 15 mA
- Excellent ac performance for low power
  - 800  $\mu$ A maximum quiescent current per amplifier
  - Unity-gain bandwidth: 1.8 MHz
  - Slew rate of 3 V/ $\mu$ s
- Good dc performance
  - 800  $\mu$ V maximum input offset voltage
  - 2  $\mu$ V/ $^{\circ}$ C typical offset voltage drift
  - 25 pA maximum input bias current
- Low noise
  - 13 nV/ $\sqrt{\text{Hz}}$  @ 10 kHz
  - No phase inversion

### ENHANCED PRODUCT FEATURES

- Supports defense and aerospace applications (AQEC standard)
- Military temperature range ( $-55^{\circ}\text{C}$  to  $+125^{\circ}\text{C}$ )
- Controlled manufacturing baseline
- One assembly/test site
- One fabrication site
- Enhanced product change notification
- Qualification data available on request

### APPLICATIONS

- Photodiode preamps
- Active filters
- 12-bit to 14-bit data acquisition systems
- Low power references and regulators

### CONNECTION DIAGRAM

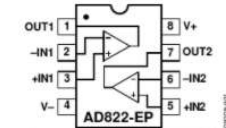


Figure 1. 8-Lead SOIC\_N (R Suffix)

### GENERAL DESCRIPTION

The AD822-EP is a dual precision, low power FET input op amp that can operate from a single supply of 5 V to 30 V or dual supplies of  $\pm 2.5$  V to  $\pm 15$  V. It has true single-supply capability with an input voltage range extending below the negative rail, allowing the AD822 to accommodate input signals below ground in the single-supply mode. Output voltage swing extends to within 10 mV of each rail, providing the maximum output dynamic range.

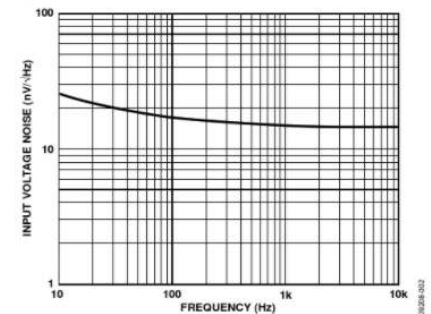
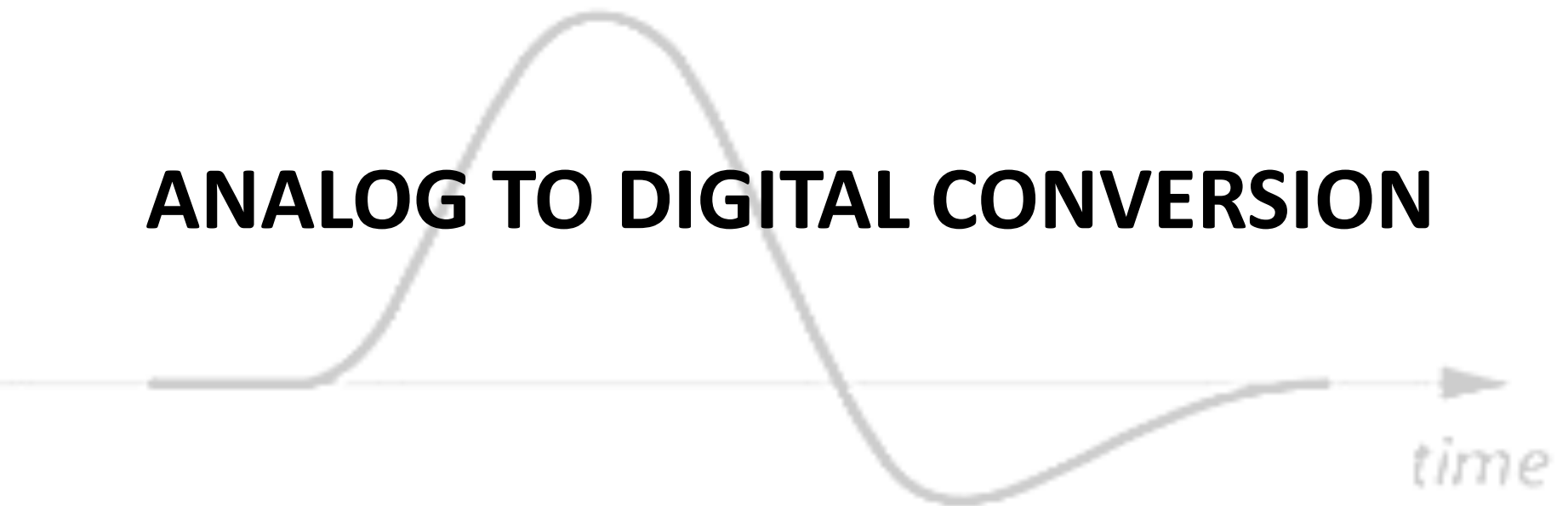


Figure 2. Input Voltage Noise vs. Frequency

Offset voltage of 800  $\mu$ V maximum, offset voltage drift of 2  $\mu$ V/ $^{\circ}$ C, input bias currents below 25 pA, and low input voltage noise provide dc precision with source impedances up to a gigaohm. The 1.8 MHz unity-gain bandwidth,  $-93$  dB THD at 10 kHz, and 3 V/ $\mu$ s slew rate are provided with a low supply current of 800  $\mu$ A per amplifier.



# ANALOG TO DIGITAL CONVERSION

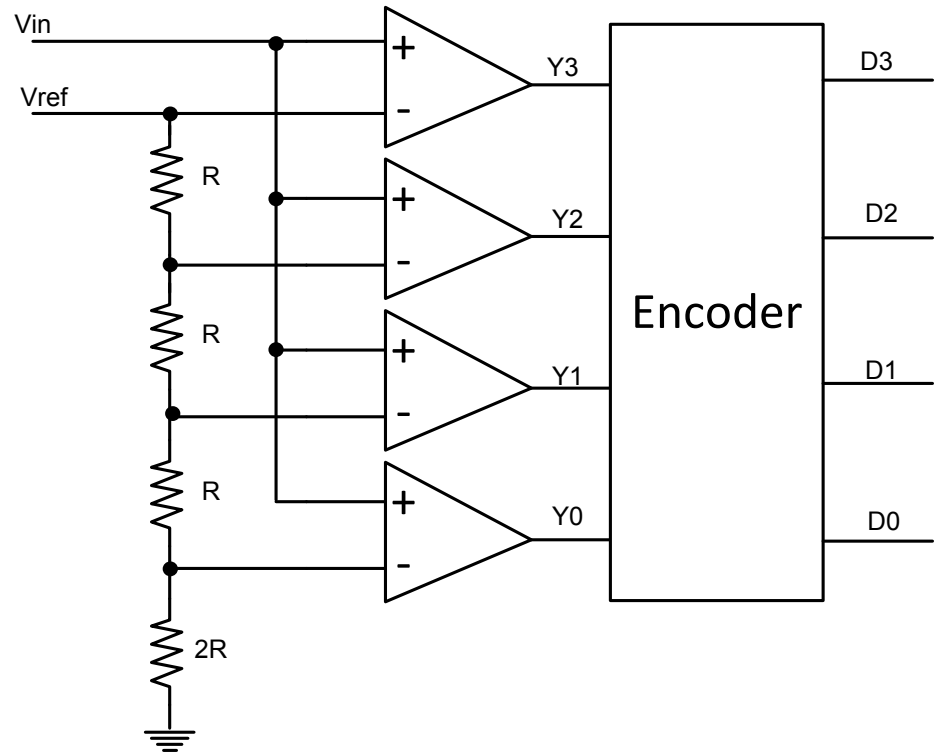


# ADC circuits

- Characteristic of ADCs
  - Resolution
  - Conversion speed
- Type of ADC circuit in the present
  - Flash ADC
  - Pipelined ADC
  - Successive Approximation Register
  - Dual-slope convertor
  - Sigma-Delta converter

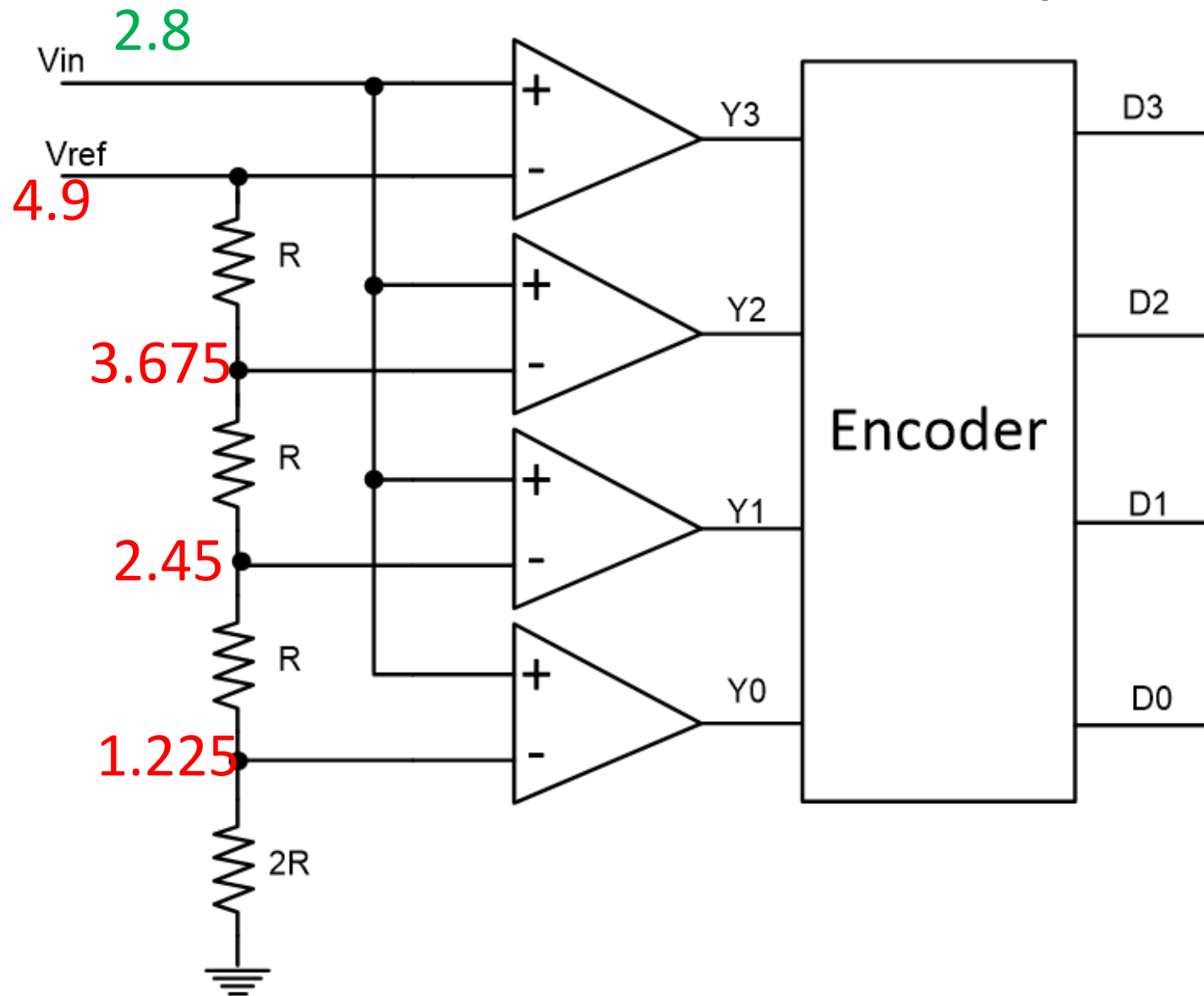
# Flash ADC

- Serie resistor for voltage dividing to create voltage references
- Compare the voltage reference with an input voltage with Opamps.
- Use encoder circuit convert to BCD data.
- Strength
  - Very fast conversion
- Weakness
  - Number of comparator require  $2^{N-1}$

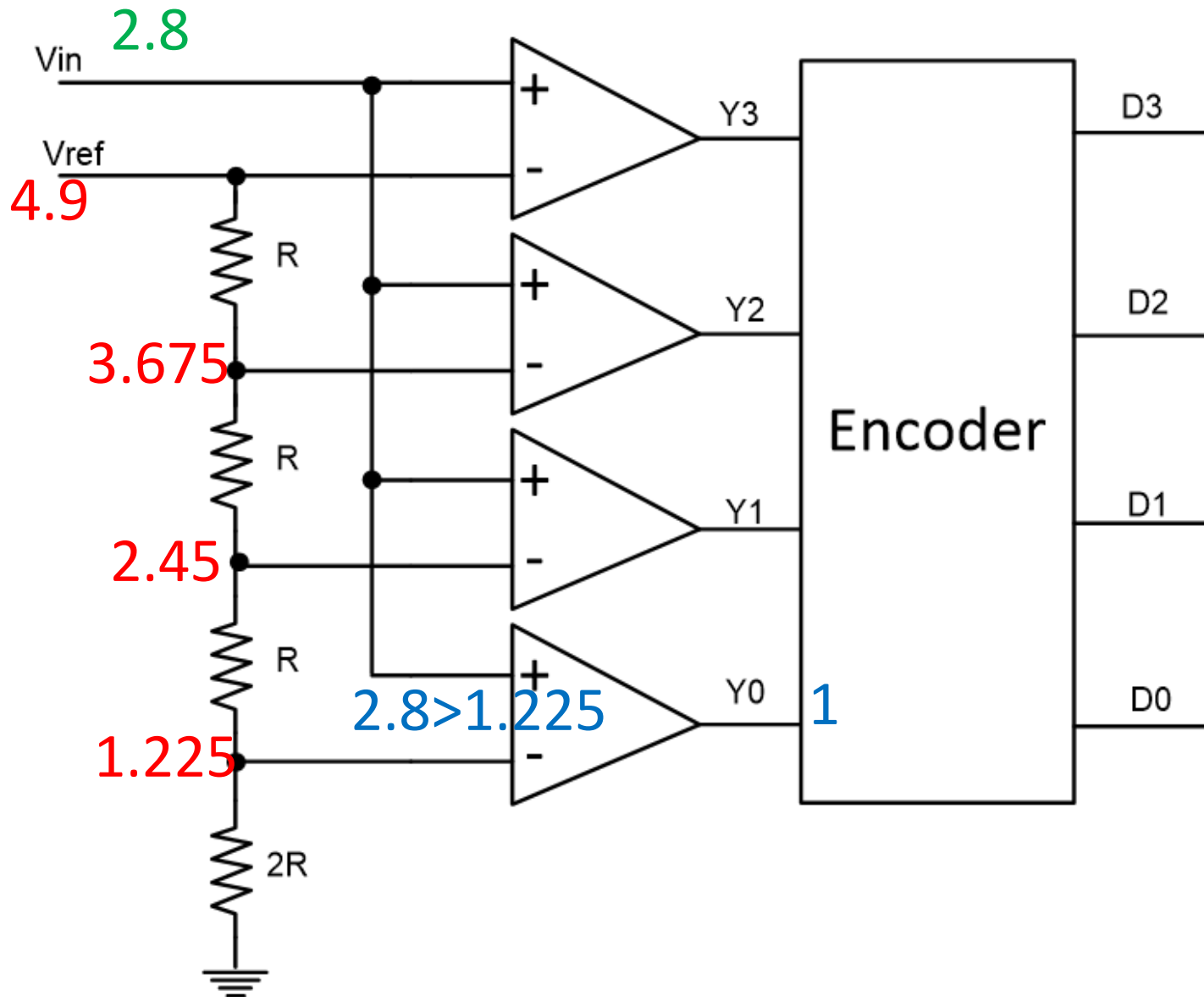




# Flash ADC: Example



# Flash ADC: Example



# Flash ADC: Encoder design

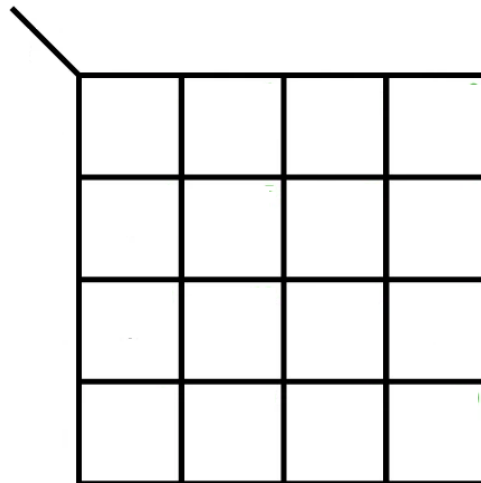
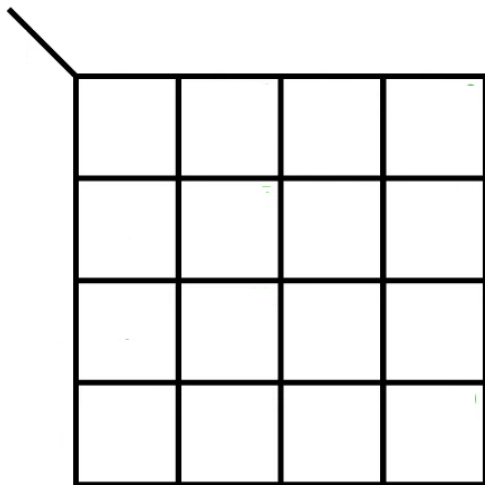
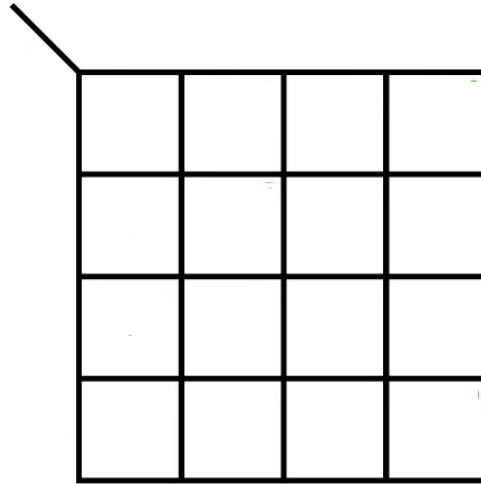
| No. | Y3 | Y2 | Y1 | Y0 | D2 | D1 | D0 |
|-----|----|----|----|----|----|----|----|
| 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 1   | 0  | 0  | 0  | 1  | 0  | 0  | 1  |
| 2   | 0  | 0  | 1  | 1  | 0  | 1  | 0  |
| 3   | 0  | 1  | 1  | 1  | 0  | 1  | 1  |
| 4   | 1  | 1  | 1  | 1  | 1  | 0  | 0  |
|     |    |    |    |    |    |    |    |
|     |    |    |    |    |    |    |    |
|     |    |    |    |    |    |    |    |

# Encoder design (added don't care bits)

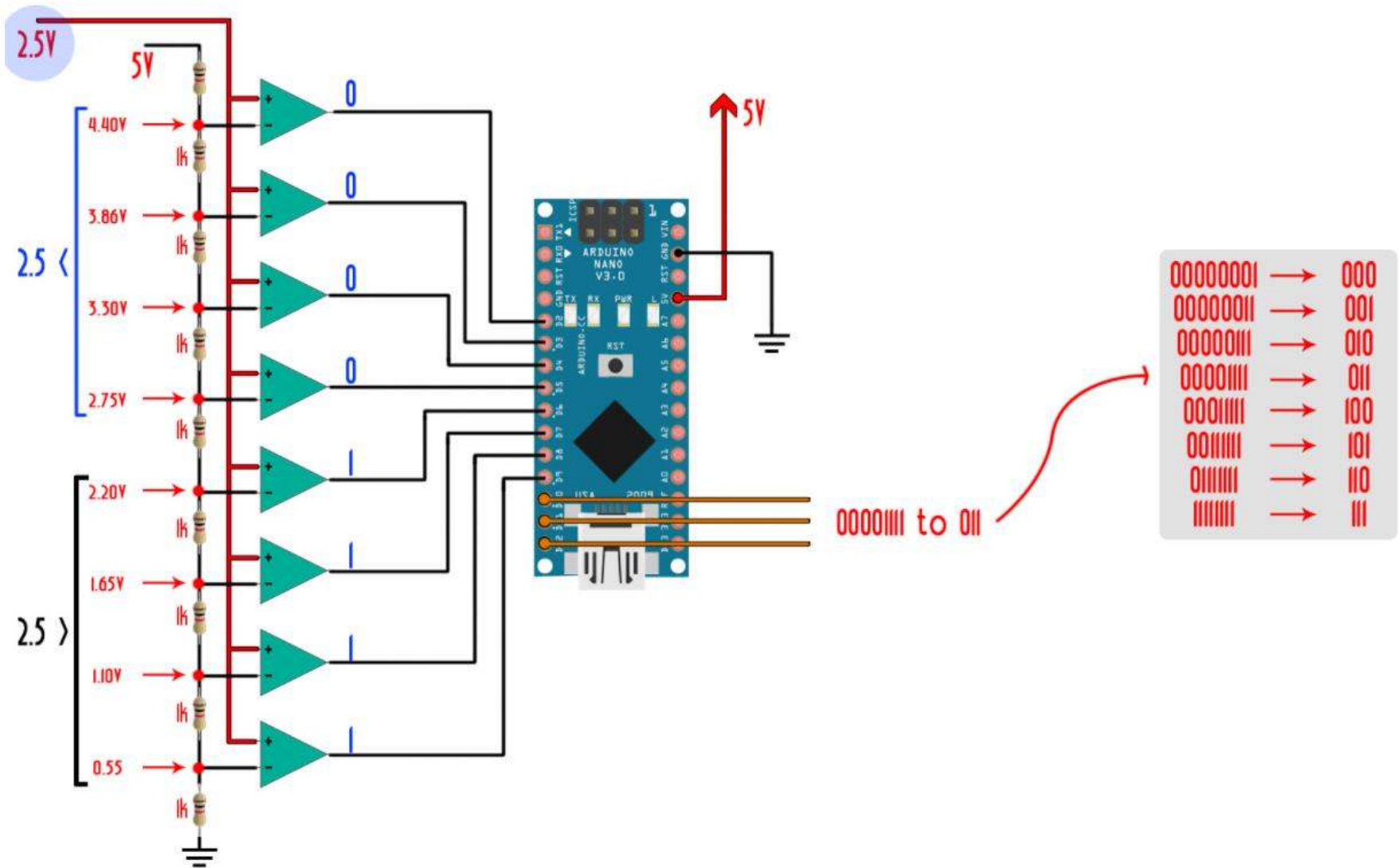
| No. | Y3 | Y2 | Y1 | Y0 | D2 | D1 | D0 |
|-----|----|----|----|----|----|----|----|
| 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 1   | 0  | 0  | 0  | 1  | 0  | 0  | 1  |
| 2   | 0  | 0  | 1  | 1  | 0  | 1  | 0  |
| 3   | 0  | 1  | 1  | 1  | 0  | 1  | 1  |
| 4   | 1  | 1  | 1  | 1  | 1  | 0  | 0  |
| 5   | 1  | X  | X  | X  | 1  | 0  | 0  |
| 6   | 1  | X  | X  | X  | 1  | 0  | 0  |
| 7   | 1  | X  | X  | X  | 1  | 0  | 0  |

# Encoder design (SOP, K-MAP)

| No. | Y3 | Y2 | Y1 | Y0 | D2 | D1 | D0 |
|-----|----|----|----|----|----|----|----|
| 0   | 0  | 0  | 0  | 0  | 0  | 0  | 0  |
| 1   | 0  | 0  | 0  | 1  | 0  | 0  | 1  |
| 2   | 0  | 0  | 1  | 1  | 0  | 1  | 0  |
| 3   | 0  | 1  | 1  | 1  | 0  | 1  | 1  |
| 4   | 1  | 1  | 1  | 1  | 1  | 0  | 0  |
| 5   | 1  | X  | X  | X  | 1  | 0  | 0  |
| 6   | 1  | X  | X  | X  | 1  | 0  | 0  |
| 7   | 1  | X  | X  | X  | 1  | 0  | 0  |



# Flash ADC applied Arduino decoder

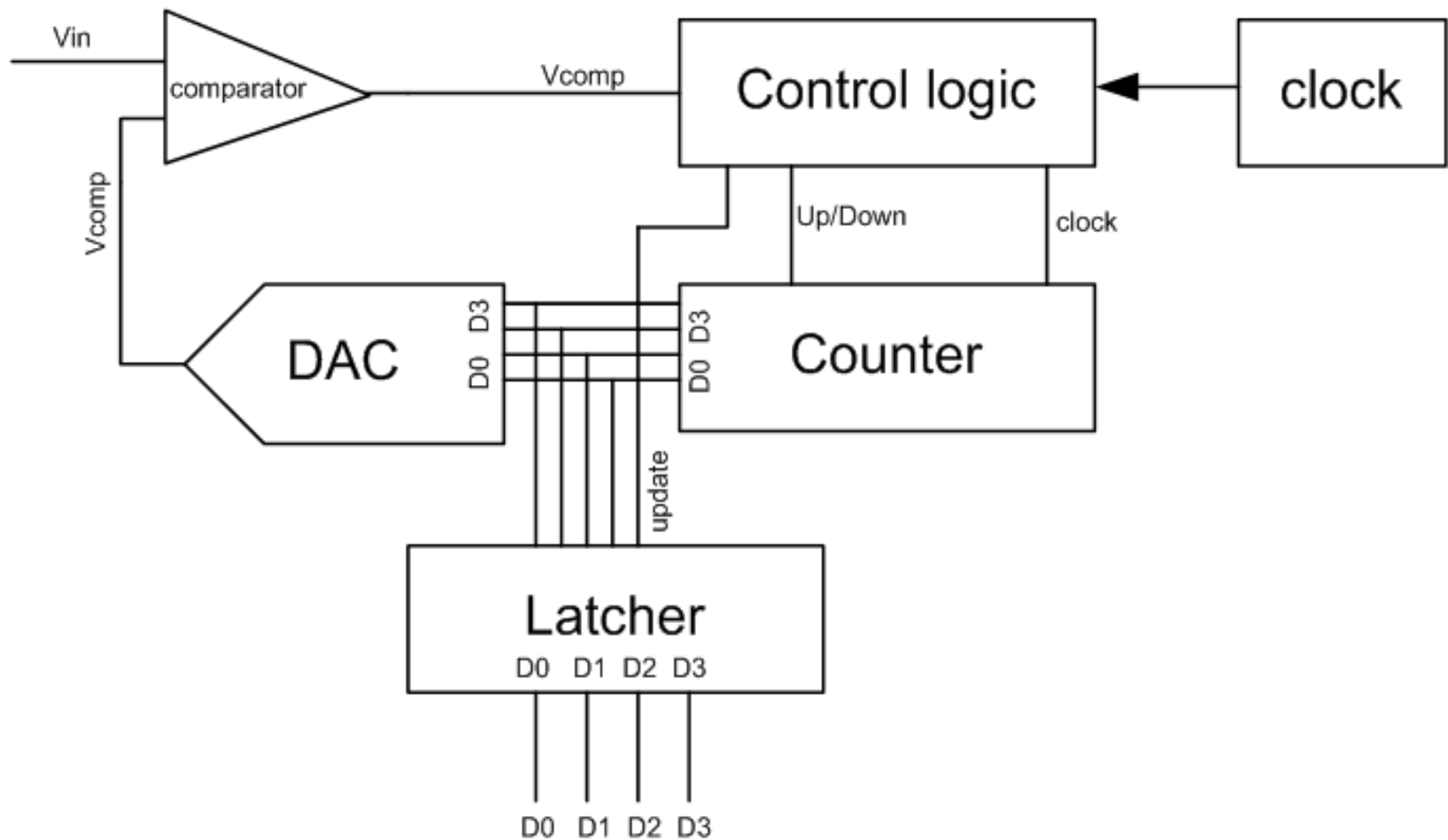


# Successive Approximation Register

## Key features

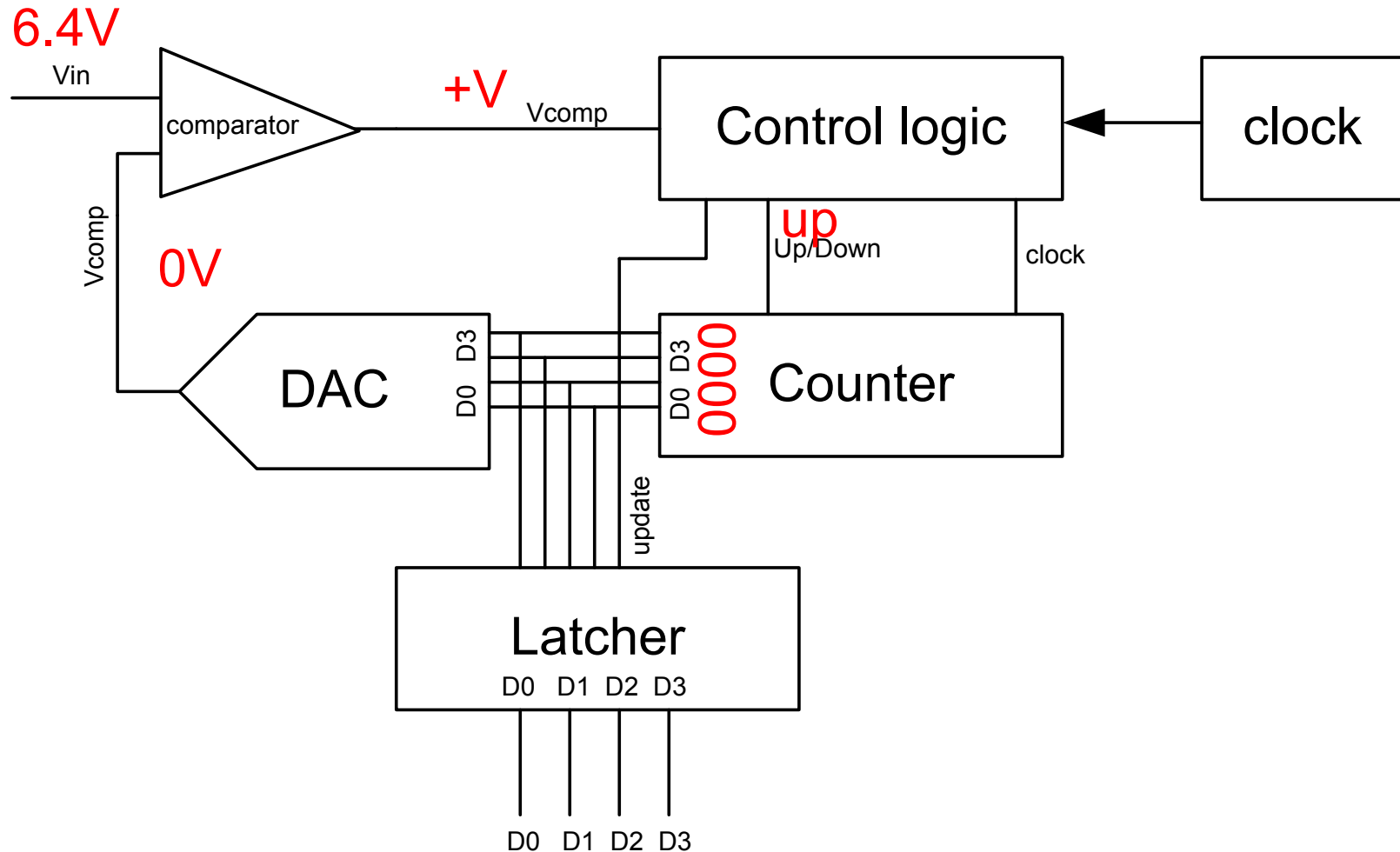
- Low energy.
- Interesting for digital circuit studying cause of having a counter, DAC, latch, and a comparator build to the system.

# SAR-ADC

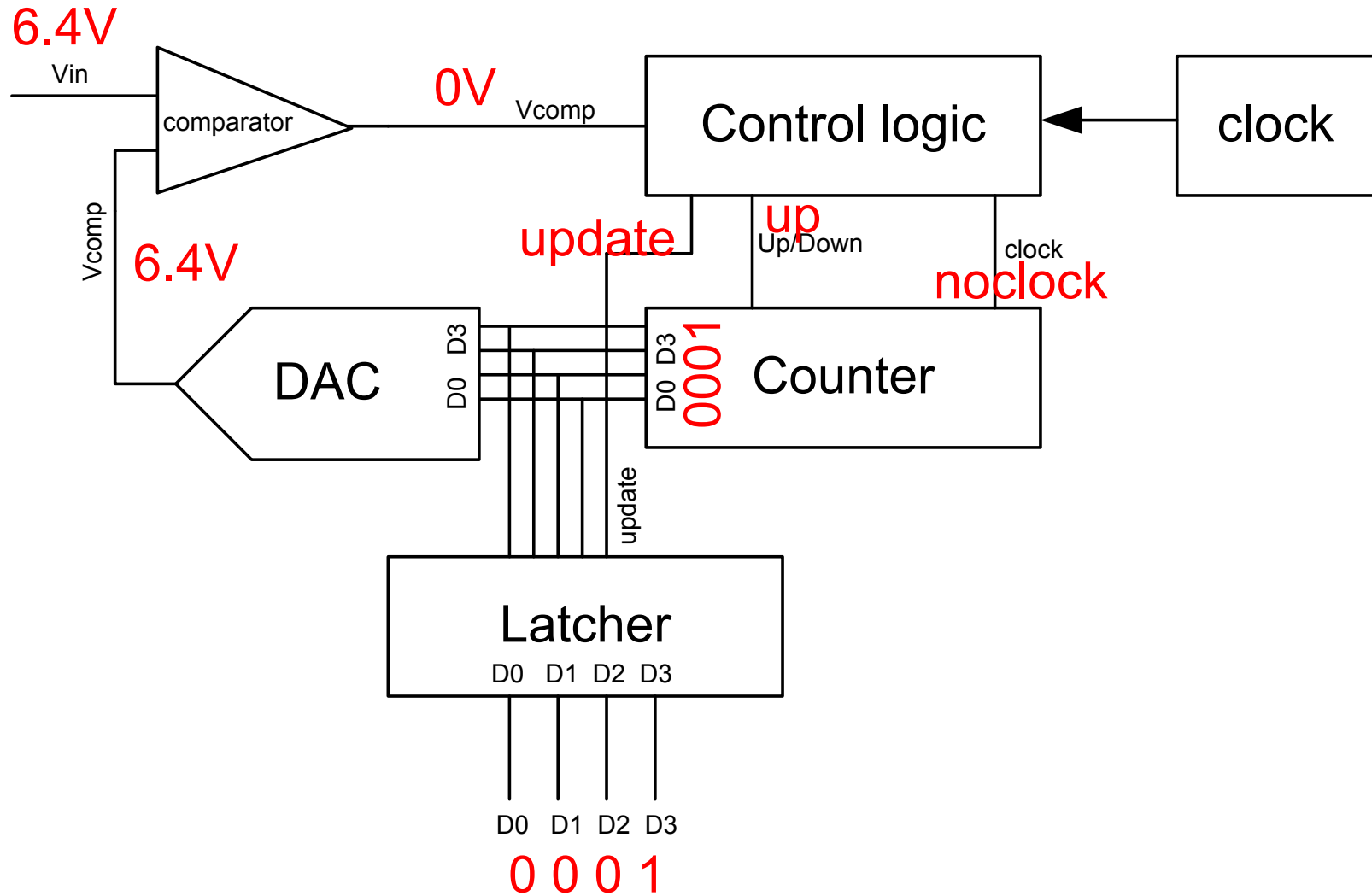




# SAR-ADC: Example running



# SAR-ADC: Example getting data



# References

| Content                                 | Links                                                                                                                                                                                                     |
|-----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Tutorial digital circuit                | <a href="https://www.tutorialspoint.com/digital_circuits/digital_circuits_conversion_of_flip_flops.htm">https://www.tutorialspoint.com/digital_circuits/digital_circuits_conversion_of_flip_flops.htm</a> |
| K-map online                            | <a href="http://www.32x8.com/index.html">http://www.32x8.com/index.html</a>                                                                                                                               |
| Encoder circuit design                  | <a href="https://www.geeksforgeeks.org/encoder-in-digital-logic/">https://www.geeksforgeeks.org/encoder-in-digital-logic/</a>                                                                             |
| 3bits flash ADC applied Arduino encoder | <a href="https://www.electrooobs.com/eng_circuitos_tut15_2.php">https://www.electrooobs.com/eng_circuitos_tut15_2.php</a>                                                                                 |
| Flash ADC                               | <a href="https://www.allaboutcircuits.com/textbook/digital/chpt-13/flash-adc/">https://www.allaboutcircuits.com/textbook/digital/chpt-13/flash-adc/</a>                                                   |
| Example ADC with simulation             | <a href="http://www.onmyphd.com/?p=analog.digital.converter#h3_resolution">http://www.onmyphd.com/?p=analog.digital.converter#h3_resolution</a>                                                           |